



Summer 2022 Exam

MY457

Causal Inference for Observational and Experimental Studies

Suitable for all candidates

Instructions to candidates

This paper contains four questions. Answer all four questions.

Question 1 is worth 40 marks in total, Question 2 27 marks, Question 3 12 marks, and Question 4 21 marks. The marks in brackets indicate marks available for each part of each question.

Time allowed **Reading Time:** *None*
 Writing Time: *2 hours*

You are supplied with: *No additional materials*

You may also use: *No additional materials*

Calculators: *Calculators are allowed in this exam*

1. (a) Consider a hypothetical study with twelve units and two experimental groups: treatment and control. The table below shows for the units in this study the values of the potential outcomes Y_1 and Y_0 (for treatment and control, respectively), the treatment indicator D (with a value of 1 indicating that a unit was assigned to treatment and a value of 0 indicating a unit was assigned to control), and a binary covariate X with values 0 and 1 (*note: X is used only in 1(b), not in 1(a)*).

Use these data to answer questions 1(a) and 1(b) below, doing calculations by hand (i.e. with a calculator) where needed.

Unit	Y_0	Y_1	D	X
1	28	28	0	0
2	16	20	0	0
3	26	26	1	0
4	22	26	1	0
5	5	5	1	0
6	12	12	0	0
7	25	29	0	1
8	19	24	1	1
9	15	20	0	1
10	16	21	0	1
11	26	30	1	1
12	14	17	1	1

- i. Calculate the true average treatment effect (ATE) from this study based on the potential outcomes. [3 marks]
- ii. What variables are included in the dataset of observed data from this study? Write down the full data matrix of the values of these observed variables. [3 marks]
- iii. Calculate the *estimated* average treatment effect based on the observed outcomes here. [3 marks]
- iv. Calculate the average values of Y_0 separately among the treated units and among the control units in this study, and also calculate the average values of Y_1 separately among the treated units and the control units. [3 marks]
- v. Considering your findings from 1(a)iv, do you think that the estimate you calculated in 1(a)iii is an appropriate estimate of the true ATE here? Why or why not? [6 marks]

(b) Here we continue to consider the data that were introduced in 1(a), but now also considering the role of the covariate X .

- i. Calculate the estimated propensity scores for being assigned to the treatment group, conditional on the two values of X . [3 marks]
- ii. Calculate the average values of Y_0 and of Y_1 among the units with $X = 0$, and the difference between these averages. Calculate similarly the average values of Y_0 and of Y_1 among the units with $X = 1$, and the difference between these averages. [3 marks]
- iii. Explain what you learn from the results in 1(b)i and 1(b)ii. Based on these findings, how would you suggest using or not using X here, as part of estimation of causal effects of the treatment on the outcome? In what ways would this be essential and/or useful here? [6 marks]

(c) Consider a general situation where we consider the potential outcomes Y_0 and Y_1 for an outcome Y , a treatment indicator D and a covariate X with possible values $x = 1, \dots, H$. In the notation below, $E(\cdot|\cdot)$ denotes a conditional expected value (average) over units in a population, and $p(X = x)$ denotes proportions of the different values of X in the population. Consider the following derivation:

$$\sum_{x=1}^H [E(Y|D = 1, X = x) - E(Y|D = 0, X = x)] p(X = x) \quad (1)$$

$$= \sum_{x=1}^H [E(Y_1|D = 1, X = x) - E(Y_0|D = 0, X = x)] p(X = x) \quad (2)$$

$$= \sum_{x=1}^H [E(Y_1|X = x) - E(Y_0|X = x)] p(X = x) \quad (3)$$

$$= E(Y_1) - E(Y_0) \quad (4)$$

- i. If this holds, we can use the expression on line (1) to obtain an estimator of the causal effect on line (4). However, this result requires a crucial assumption, which allows us to move from line (2) to line (3). Explain, in your own words or in appropriate notation, what this assumption is. [5 marks]
- ii. Explain what is meant by a *stratified randomized experiment*, and explain why the assumption discussed in 1(c)i is satisfied in such an experiment. [5 marks]

2. Consider a hypothetical study with twelve units and two experimental groups: treatment and control. The table below shows for the units in this study the values of the potential outcomes Y_1 and Y_0 (for treatment and control, respectively), the treatment indicator D (with a value of 1 indicating that a unit was assigned to treatment and a value of 0 indicating a unit was assigned to control), and a continuous covariate X . Use these data to answer the questions below.

Unit	Y_0	Y_1	D	X
1	5.3	8.6	0	1.4
2	5.4	9.7	1	2.4
3	4.9	9.3	1	2.7
4	4.8	10.5	0	3.1
5	5.3	10.0	1	3.4
6	4.1	10.3	1	3.5
7	5.3	11.2	0	3.6
8	4.9	10.7	1	3.6
9	4.9	11.2	0	3.8
10	4.5	11.5	0	3.8
11	4.9	15.2	1	8.5
12	5.9	15.5	0	9.3

- (a) Consider first matching estimators of the average treatment effect on the treated (ATT) for this study, where we match units based on their observed values of X . Here it is convenient to display these values in ascending order within the control group and within the treatment group, like this:

$D = 0$	$D = 1$
1.4	2.4
3.1	2.7
3.6	3.4
3.8	3.5
3.8	3.6
9.3	8.5

- Suppose that we carry out 1 : 1 nearest-neighbour matching *with replacement* for these data. Which treated units are matched with which control units in this exercise? Identify the matches by writing down the table above and drawing a line between each pair of values of X that are matched. [3 marks]
- Suppose now that we carry out 1 : 1 nearest-neighbour matching *without replacement* for these data. Identify, in a similar way as in 2(a)i, which treated units are matched with which control units in this exercise. [3 marks]
 - Note:* To fix the order in which the treated units are considered, begin with the unit for which the closest possible match is available. Then in each subsequent step match the unit for which the best available match is closest.
- Calculate the matching estimates of ATT from the matched samples in 2(a)i and 2(a)ii. [4 marks]
- In general, what are the relative advantages and disadvantages of matching with replacement, compared to matching without replacement? [5 marks]

- (b) We now consider regression-based estimation of the average treatment effect (ATE) for this study. We fit the following linear regression model

$$Y_i = \alpha + \tau D_i + \beta X_i + \epsilon_i \quad (5)$$

where $i = 1, \dots, n = 12$ denote the units, and ϵ_i the model residuals. The least squares estimate of τ for these data is $\hat{\tau} = 4.64$.

- i. In general, explain which assumption of matching estimators is not required for regression-based estimation of causal effects. In contrast, what assumption is required by regression-based estimators but not by matching estimators? What key assumption is required by both of them? [6 marks]
 - ii. Considering the observed data in this study, do you think that the estimate of τ that is obtained by fitting the model specified as in (5) is an appropriate estimate of the true ATE here? Why or why not? [6 marks]
3. Briefly describe, in your own words and/or appropriate notation, the key elements, basic ideas and assumptions of the sharp regression discontinuity design. Explain what causal effects can be estimated from it, and how they are estimated. Discuss the strengths and weaknesses of this design. [12 marks]

4. Consider an experiment about a voter mobilization campaign in the United States. An organization wanted to find out whether face-to-face contacts with voters by non-partisan advocacy groups can increase voter turnout in a subsequent election. Individuals were randomly assigned to either receive a face-to-face contact via a surprise visit by a member of the organization (the treatment), or to a no-contact control group. Not all individuals assigned to the treatment group actually received the face-to-face contacts in their homes. The following table presents a summary of the data:

	Group size	Proportion who voted
Assigned to treatment		
Received face-to-face contact	10,836	.62
Was not contacted	2,482	.54
Assigned to control		
Received face-to-face contact	0	.52
Was not contacted	11,637	.35

Assume that you want to use these data in an instrumental variables analysis in which voting is the outcome (Y), receiving face-to-face contact is the treatment (D), and being assigned to receive face-to-face contact is the instrument (Z).

- Using the data in the table, calculate the estimated intention-to-treat (ITT) effect of being assigned to the treatment group on voting. That is, calculate the difference in the proportion who voted between those who were assigned to the treatment group and those who were assigned to the control group. This is also the ordinary least squares estimate of γ_2 in the regression model $Y = \gamma_1 + \gamma_2 Z + \epsilon$. [3 marks]
- Using the data in the table, calculate the estimated ITT effect of being assigned to the treatment group on receiving the face-to-face contact. [3 marks]
- Calculate the instrumental variables estimate of the local average treatment effect (LATE) of the treatment on the outcome for the compliers. [3 marks]
- Explain the assumptions that a valid instrument would have to fulfill in the context of this particular study. For each assumption, evaluate whether randomized treatment assignment is a valid instrument in the context of this study. Explain why or why not. [12 marks]